INTRODUCTION

The following is a compendium of notes on standard practices and drawing symbols used in the General Site/Balance of Plant electrical installation.

Areas addressed include:

- General Notes
- Power Level Classification/Separation
- Conduit/Duct Bank Notes
- Buried Power Cable Notes
- Grounding Notes
- Lighting Notes
- Electrical Symbols (Typical)
- Lighting Symbols (Typical)
- Installation Drawing Symbols (Typical)

Installation of Gas Turbine-Generator supplied control cables and conduit transitions are covered in separate EM articles.

Installation of the Turbine-Generator supplied auxiliary transformer and starting motor circuits and generator bus bars is covered, as appropriate, in the specific unit installation drawings.

OTHER ISSUES

- Review Equipment Schedule, Installation And Quality Problems
- Customer Evaluation Of The Job
- Inspection Intervals
- The First Combustion Inspection Should Be Identified By The Customer At This Meeting.
- Review Spare Parts

- A List Of Spare Parts Should Be Made Available At This Meeting.
- Other

RECOMMENDED DOCUMENTS

TDI Contract - District.

The meeting will be conducted in such a manner that all outstanding items are reviewed with the customer. Agreements will be reached & documented concerning the following.

- Completion of the punch list items
- Submittal of the Installation/ Start-up Report
- Completion of the Marked-up "AS-BUILT" Drawings
- Warranty Start Date

After the areas listed above have been addressed, the PGSD obligation will be concluded, provided the remaining punch list items are not associated with the TDI scope of responsibility.

OTHER CONSIDERATIONS

- After the customer meeting, PGSD will publish the finalized installation punch list, including a schedule and meeting minutes documenting agreements, action items, warranty start date, and commercial operating date.
- The status of the outstanding punch list items shall be issued monthly by PGSD and distributed to the installation support engineer until all the items are resolved.
- The Product Department installation and/or service engineers will have the option of attending and participating in post-installation meetings.

POWER LEVEL CLASSIFICATIONS

Level	Definition	Typical Applications	Installation Practices
1 (Low Level)	a. Analog signals 0-50 VDC <60 MA b. Digital signals 0-12 VDC	a. ACOM, DCOM, CCOM, SCOM b. Feed buses for analog or digital conductors c. Logic input and/or output signals d. Transducer outputs e. Temp. indicators f. RTDs	a. Twisted-shielded pair conductors must be contained in separate enclosure of cable and isolated from other higher level signals.
1S (Special Low Level)	a. Millivolt Range b. Extremely low currents	a. Thermocouples b. Thermocouple extension wire	a. Thermocouple extension wires (can be run with other Level 1 signals if independently shielded). b. Extension wires must be twisted pairs with at least one shield covering all pairs in a single cable.
2 (Medium Level)	a. Analog signal >50 VDC <28 VAC ripple b. Digital signal to 18 VDC <0.6 VA	None	
3 (High Level)	a. Analog signals >50 VDC >28 VAC ripple b. AC feeders (120 VAC) <20 AMPS c. DC feeders (120 VAC) <20 AMPS d. Switching signals 0-	a. AC lighting and convenience outlets b. AC Instrumentation power c. DC relay and solenoid power d. Optical isolator power	a. Must be run in enclosure or cable with other Level 3 signals. Generally must not be run with higher level signals.

	125 VDC <20 AMPS	e. Input filters switching signals	
3S (Special High Level)	a. 300 VDC (saw tooth to DC)	a. Flame detectors only	a. Must be run in separate enclosure or cable with other flame detector signals only. b. When terminating in junction box, it can enter box with other levels for short distance.(less than 2 ft.) c. Terminal board used for this level must be separate from all other level boards. d. Cable must be low capacitance, twisted shielded pairs. (Capacitance 55 PFD/ft, maximum of 850 ft.)
4 (Power Level – Low Voltage)	a. AC/DC equipment 0-600V/20-800 AMPS	a. Motor armature loops b. Generator armature loops c. Power transformers (> 5 kVA) d. Machine fields (Motor or generator fields) e. Motor control contacts f. Heaters	a. Generally run in separate enclosure or cable with other Level 4 type signals. May be run with Level 3 if Level 3 limits apply.
5 (Power Level –	a. AC/DC equipment 4 kV and above	a. Limit-amp motor starter	a. Generally run in separate enclosures

Medium Voltage)		b. Starting motor c. Generator step-up transformer	away from Level 1 and 2 circuits. Note: Separate raceways are required for cables with different insulation ratings, i.e., 5 kV, 15 kV, 35 kV, etc.

POWER LEVEL SEPARATION

Minimum Distance Between Conduit Runs

12"	Level 5	To Level 1 or 1S or 2
12"	Level 4	To Level 1 or 1S
9"	Level 4	To Level 2
3"	Level 3 or 3S	To Level 1, 1S or 2

Minimum Distance Between Cables Run

84"	Level 5	To Level 1, 1S or 2
29"	Level 5	To Level 3 or 3S
28"	Level 4	To Level 1, 1S or 2
20"	Level 4	To Level 3 or 3S
6"	Level 3 or 3S	To Level 1, 1S or 2
2"	Level 2	To Level 1 or 1S

Minimum Distance Between Cable & Conduit Runs

48"	Level 5	To Levels 1, 1S or 2
20"	Level 4	To Levels 1, 1S or 2
18"	Level 5	To Levels 3 or 3S
12"	Level 4	To Level 3 or 3S
4"	Level 3 or 3S	To Levels 1, 1S or 2
2"	Level 2	To Levels 1 or 1S

CONDUIT/DUCT BANK NOTES

1. Conduit which have been cut are to be reamed to remove rough edges and burrs. Metallic conduit is normally threaded at both ends, unless otherwise noted.

2. Generally conduits are to project 3 inches above the finished floor or grade, and 3 inches beyond the finished wall, except for conduits entering switchgear from below which should not extend more than 1 inch above finished floor.

3. Buried, intermediate or rigid metallic conduit are to be at least 18 inches below grade, unless protected all around with corrosion resistant material or encased in a 2 inches concrete envelope.

4. Buried, non-metallic conduit are to be a minimum of 18 inches below grade, measured from the top of the conduit or 12 inches below grade if a 2 inch concrete pad (or equivalent physical protection) is placed in the trench over the underground installation or just below grade if protected under a 4-inch thick concrete slab which extends at least 6 inches beyond the conduit in either direction.

5. Buried, non-metallic conduits, used for installation of 600V or higher voltage service in all cases, are to be enclosed in a concrete envelope of at least 2 inches thick.

6. Under areas of heavy vehicular traffic, the depth for conduit should be 2 feet or greater and are to be encased in reinforced concrete.

7. It is recommended that when concrete is used to encase conduit that it be identified by red coloring, (e.g., red dye placed on or in the concrete).

8. Flexible conduit is normally limited to a length of 18 inches with a maximum length of 36 inches.

9. Intermediate or rigid metallic conduit 1 1/2 inch diameter and smaller can be bent in the field. Use of factory fabricated elbows is recommended on all intermediate or rigid metallic conduit greater than 1 1/2 inch diameter.

10. Should additional conduit be added to the main centerline foundation, reference should be made to MLI 0323 for minimum depth required for conduit running under the exhaust plenum. A minimum depth of 24 inches is typical to prevent overheating.

11. Before pouring foundation verify that embedment restraints are not touching conduit, (to prevent direct conduction of heat to conduit) and that conduits are capped at both ends.

12. Installed conduit is never used as a ground conductor for welding.

BURIED POWER CABLE NOTES

1. A sand cushion of at least 6 inches thickness should be compacted in the bottom of the trench before laying the first row of cable.

2. The cable should be laid with a slight snaking to prevent overstressing upon earth settling and thermal movement during load cycling. Power cables of the same level may require spacing to allow for needed heat dissipation, look for specific recommendation on associated drawings.

3. The sand which is backfilled between the lower and upper rows of cable should be well compacted before laying the upper rows in order to minimize its settling. Care must be taken in order not to damage the lower cables.

4. After carefully compacting the 6 inches of sand covering the top row of cables, creosotecoated planking, unglazed tile, or a 4-inch concrete slab should be placed above the cable runs for mechanical protection as well as to warn workmen of its presence.

5. Suitable above ground markers are to be installed to indicate the cable run.

6. Refer to the applicable construction drawings for the specific cable routing as well as depth and spacing dimensions.

7. The orientation and spacing of the cables shall strictly follow the arrangement shown on the construction drawings.

GROUNDING NOTES

1. All non-current carrying metallic objects such as switchyard structures, fencing, building steel, equipment enclosures, pullboxes, support structures and other frame work, shielded cable sheaths, metal conduit, pipe, etc. are to be grounded to a common earth potential whether or not specific details of such appear on the construction drawings. Typically two #4/0 conductors are used.

2. Building steel and fencing are to be grounded at intervals not exceeding 70 feet - 0 inches unless indicated otherwise. Normally small skids are grounded at two opposing corners. Turbine-Generators generally provide 2 grounding lugs per compartment.

3. Metallic conduits are to be connected to the ground grid at one end only. The connection to the ground grid must utilize a ground cable.

4. All equipment ground connections must be electrically continuous from the equipment to the ground grid.

5. All equipment and structures are to be grounded utilizing a visible ground connection at the equipment.

6. The equipment grounding conductor may be a metallic raceway such as steel conduit or tray or a cable. Equipment ground cable may be bare or insulated. If it is insulated, it must have a green colored insulation. If green cable is unavailable the cable used must be colored green with tape or paint, or stripped bare over the entire portion that is visible.

7. All grounding grid cable must be bare and must be buried 18 inches below grade unless indicated otherwise; it must be laid slack to prevent damage due to soil settlement.

8. All grounding grid cable riser "pigtails", for connection to a structure, should extend a minimum of 1 foot - 0 inches above the base of the structure.

9. All buried ground grid connections are to be made utilizing an approved compression type fitting or by an approved exothermic process, such as brazing or soldering.

10. No ground connections to be buried should be covered over prior to the consent of the project engineer/manager of his appointed representative.

11. All above grade ground connections are to be made utilizing approved connectors with non-corrosive bolts, nuts and washers unless indicated otherwise.

12. All ground cable connections are to be made after the connection contact surface is thoroughly cleaned of dirt and oil to ensure a good low resistance connection.

13. Ground cables are not to be run through metal conduit or through metal barriers unless all phases or polarities are included in the conduit or unless the ground cable is securely bonded at both ends of the pipe, conduit or metal barrier.

14. Conduit sleeves for ground cable penetrations are to be of a non-metallic material.

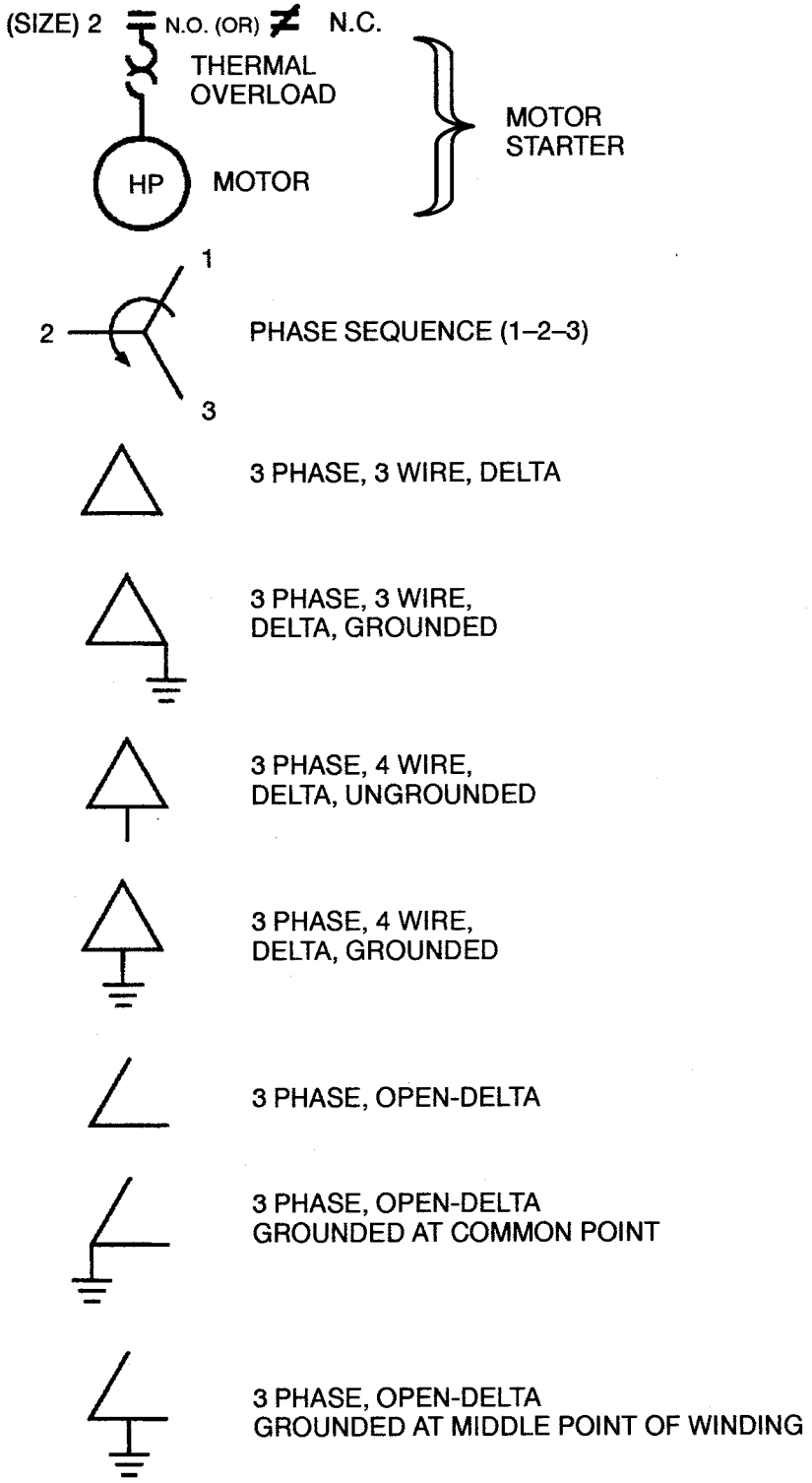
15. Typically the grounding system should have a measured nominal resistance, with respect to remote earth, of 2 Ohms or less. If this resistance is not attained by the system shown on the construction drawings, advise the project engineer/manager for further direction.

LIGHTING NOTES

1. All fixture locations shown are approximate and are to be adjustedin the field to avoid interferences with piping, tray, conduits and supports. Lighting installations should be scheduled to occur after the installation of piping, conduit and tray work to minimize or eliminate rework due to interferences.

2. Lighting cable is to be a single conductor not smaller than #12 AWG, unless otherwise noted.

ELECTRICAL SYMBOLS





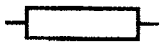
3 PHASE, WYE OR STAR
UNGROUNDED



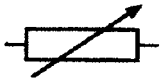
3 PHASE, WYE,
GROUNDED NEUTRAL



3 PHASE, ZIGZAG
UNGROUNDED



RESISTOR



RESISTOR – ADJUSTABLE



INSTRUMENT OR
RELAY SHUNT



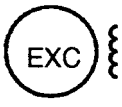
CAPACITOR



BATTERY



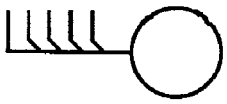
POTHEAD



EXCITER



RECTIFIER



GROUPING OF LEADS



GROUND



CROSSING OF CONDUCTORS



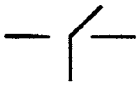
JUNCTION OF CONDUCTORS



DIRECTION OF FLOW SIGNAL



SINGLE THROW SWITCH



DOUBLE THROW SWITCH



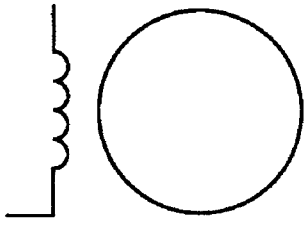
COMMUTATING FIELD



SERIES FIELD



SHUNT FIELD



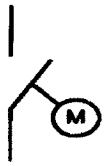
SEPARATELY EXCITED
DIRECT CURRENT
GENERATOR OR MOTOR



HEATER



SEPARABLE CONNECTORS



MOTOR OPERATED SWITCH



CURRENT TRANSFORMER



CURRENT TRANSFORMER
WITH POLARITY MARKINGS



BUSHING
CURRENT TRANSFORMER



GROUND SENSING
CURRENT TRANSFORMER



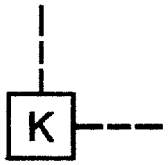
POTENTIAL TRANSFORMER



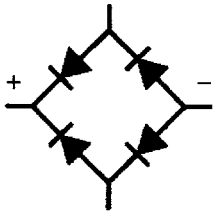
POTENTIAL TRANSFORMER
WITH POLARITY MARKINGS



FUSE



KIRK KEY INTERLOCK



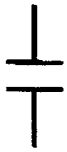
BRIDGE TYPE
RECTIFIER



TRANSFORMER WINDING
NONSATURATING
(ONE-LINE DIAGRAM)



TRANSFORMER WINDING
(ONE-LINE DIAGRAM)



CONTACT OPEN



CONTACT CLOSED



INDICATING LIGHT



LIGHTNING ARRESTER



CIRCUIT BREAKER
1500 VOLTS OR LESS
AND ALL DC CIRCUIT
BREAKERS



POWER CIRCUIT
BREAKER



CIRCUIT BREAKER
WITH THERMAL-OVERLOAD
IN ALL 3 POLES



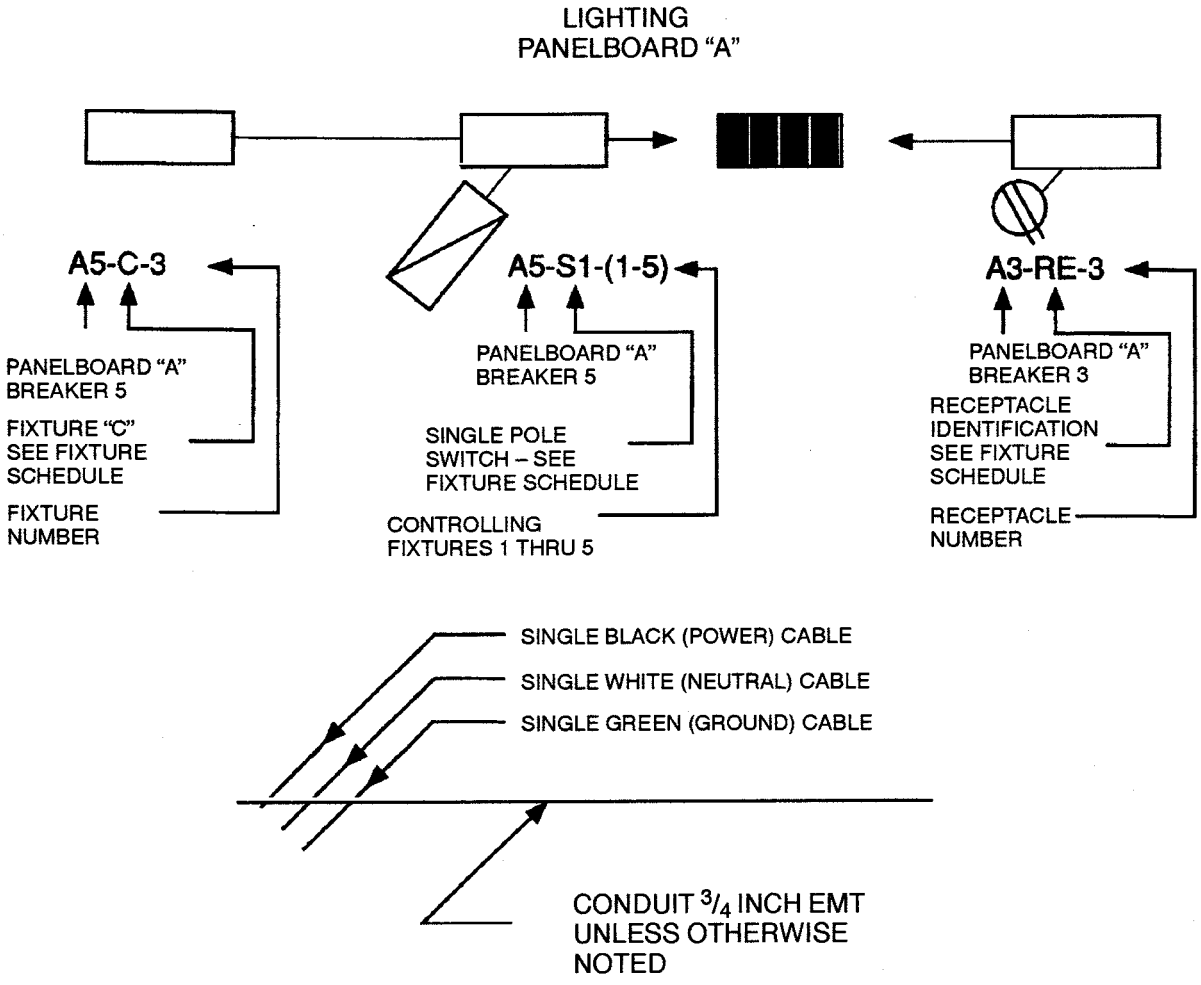
CIRCUIT BREAKER
WITH MAGNETIC OVERLOAD
IN ALL 3 POLES



CIRCUIT BREAKER
DRAWOUT TYPE



POWER CIRCUIT BREAKER
DRAWOUT TYPE





HOMERUN TO PANELBOARD (NUMBER OF CIRCUITS IS INDICATED BY THE NUMBER OF ARROWS)



CONVENIENCE OUTLET – 120 VOLT – SINGLE



CONVENIENCE OUTLET – 120 VOLT – DUPLEX



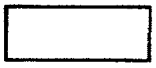
SPECIAL PURPOSE OUTLET



INCANDESCENT OR MERCURY VAPOR FIXTURE



EMERGENCY LIGHT



FLOURESCENT FIXTURE



FLOURESCENT FIXTURE WITH BUILT-IN EMERGENCY LIGHT



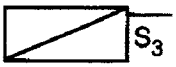
WATCHMAN LIGHTS



SINGLE POLE SWITCH



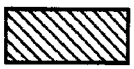
DOUBLE POLE SWITCH



THREE WAY SWITCH



FOUR WAY SWITCH



POWER PANEL



LIGHTING PANEL



ADDITION OF THE LETTER "X"
TO OUTLETS AND SWITCHES
DENOTES EQUIPMENT IS
"EXPLOSION-PROOF"

GROUND POINT

TEST POINT (GROUNDING)

CONDUIT

FLEXIBLE CONDUIT

CONDUIT TURNING UP OR TOWARD

CONDUIT TURING DOWN OR AWAY

CABLE, INSULATED OR BARE

CABLE, TURNING UP OR TOWARD

CABLE, TURNING DOWN OR AWAY

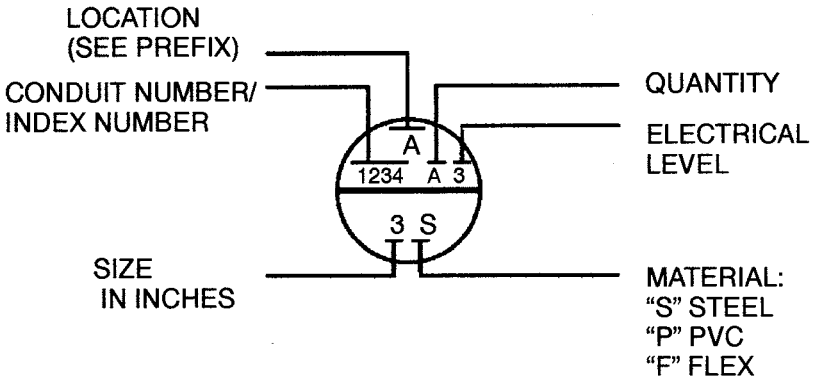
} GROUNDING

CONNECTION-GROUND CABLE TO EQUIPMENT

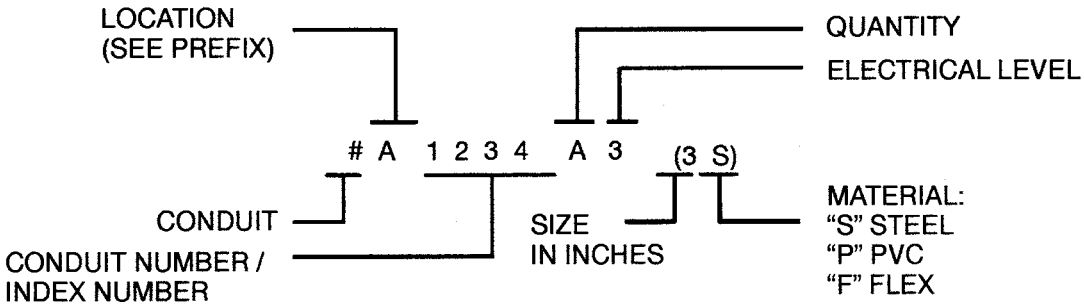
CONNECTION-CABLE TO GROUND CABLE

EXPLOSION-PROOF COUPLING

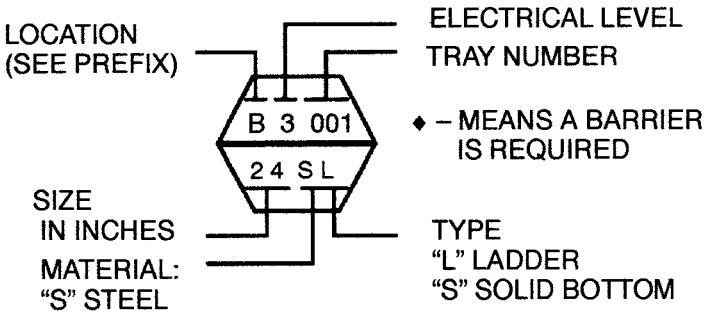
CONDUIT IDENTIFICATION SYMBOL: TEMPLATES



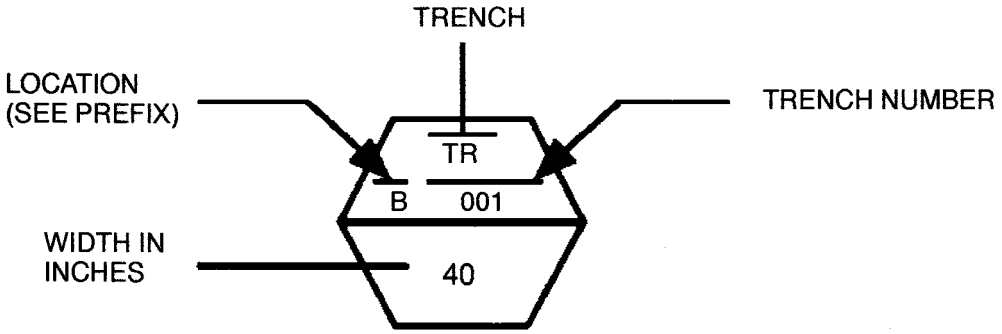
CONDUIT IDENTIFICATION: PLANS & SECTIONS



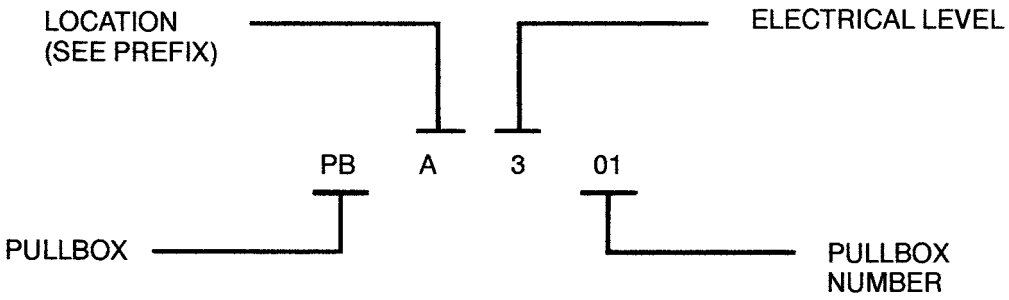
TRAY IDENTIFICATION SYMBOL

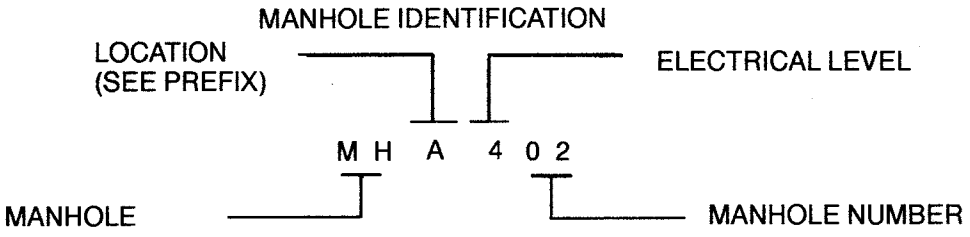


TRENCH IDENTIFICATION SYMBOL



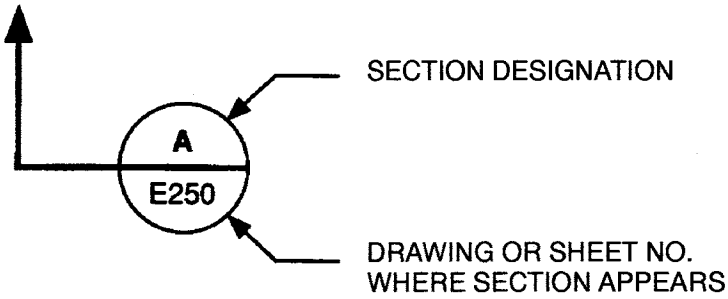
PULLBOX IDENTIFICATION



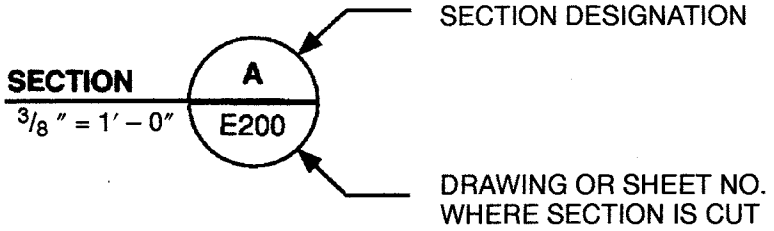


SECTION IDENTIFICATION SYMBOLS

A. PLAN DRAWING:



B. SECTION DRAWING:



DETAIL IDENTIFICATION SYMBOLS

PLAN DRAWING:

